

The effect of exoplanet atmospheres on tidal Love numbers

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Summary

Planets in close orbits around their star experience tidal forces that can lead to tidal migration, as well as atmospheric escape driven by stellar heating. Both processes can significantly alter a planet's structure and orbital evolution. Since the strength of tidal interactions depends on a planet's internal structure, the loss of its atmosphere could impact the tidal migration rate and the final architecture of planetary systems.

The strength of tidal interactions is captured by the tidal Love number k_2 , which quantifies a planet's response to tidal forces.

In this study, I used ALMA3, a numerical code for calculating Love numbers based on internal structure, to investigate how atmospheric loss influences k_2 . The results show that for planets starting with relatively small atmospheres (which extend up to a couple times the radius of the core), k_2 decreases sharply as the atmosphere is lost, but this is counterbalanced by the decreasing planetary radius, leaving the migration rate largely unchanged. In contrast, for planets with large atmospheres that dominate the overall structure, k_2 remains nearly constant while the migration rate decreases as the atmosphere shrinks. An analytical formula was developed that accurately models this behaviour and can be readily used in future tidal migration simulations.

1 Introduction

As a first order approximation, planetary systems are modeled as a system of point masses attracted through gravity. When taking into account their finite size however, gradients of the gravitational force from one body, the forcing body, across the volume of another body, the responding body, will cause the responding body to deform. These distortions cause a variety of side effects. For example, if the deformation is not static, the internal stresses and movements dissipate energy, heating up the body while extracting energy from the orbit and rotation of the bodies involved. Extracting energy from a body's orbit will cause its orbit to change, and the body to move through the system—an effect called tidal migration.

As such, tidal interactions can potentially have severe consequences on the evolution, stability and fate of planetary systems (Tremaine, 2023).

The energy dissipated in a body's interior - and thus the rate of tidal migration - depends on two main factors. First, the nature of the interaction: the relative distance and rotation rates determine how strongly the bodies perturb each other. Second, the internal structure matters. The rigidity and internal mass distribution determine how much a body will deform and how much energy is dissipated for a given perturbation (Correia and Rodríguez, 2013).

The effect of a body's internal structure on the migration rate is captured by a property called the tidal Love number, denoted by k . This macroscopic property can be derived from a known internal struc-

ture. For the simplest internal structures analytical solutions are available (Alterman et al., 1959; Peltier, 1974; Storch and Lai, 2013, 2015). For more complex structures numerical solutions are needed. An example of this is ALMA3, developed by Melini et al. (2022), which can calculate the Love numbers of a multi-layered body.

As observational methods improve, more constraints on the internal structure of exoplanets emerge. One of these constraints comes from the Love number, as it provides a direct link to the mass distribution inside the planet. Using eccentricity measurements in a multi-planet system, transit timing variations and light-curve deformation, the tidal Love numbers of five exoplanets have been determined to date (van Dijk and Miguel, 2025).

Apart from being a promising way to constrain exoplanet interior models, Love numbers are also important for the study of the evolutionary history of planetary systems. Population studies show a high occurrence rate of Jupiter-like planets close to their host star, more than can currently be explained by formation models alone. Orbital migration, including migration due to tidal interactions, of planets that formed further outward can explain the large amount of close-in planets Perryman (2018). To further study and constrain the evolutionary history of such systems, knowing the tidal Love number of involved bodies is important for further study.

Another result from population studies is the hypothesis that certain types of exoplanets in close orbits around their host stars are supposed to lose significant parts of their atmospheres (West, 2019). This hypothesis can potentially explain the observed absence of planets that are about the size of Neptune and in close orbits to their stars, also called the Hot-Neptune desert. The central idea is that these planets form further out from their star, and migrate inwards due to either tidal or disk interactions. As they get closer to their host stars, they heat up, until eventually their atmosphere becomes too hot and starts to escape the gravitational confinement of the planet. This process, called atmospheric escape, is thought to be able to strip large parts of the atmospheres from these planets. Atmospheric escape has been observed in several Hot Jupiters and Hot Neptunes and can,

especially for Hot Neptunes, significantly alter their evolution (Owen, 2019) (Bourrier et al., 2013). For example, (Owen and Wu, 2013) showed that a $20M_{\oplus}$ planet with a $12.5M_{\oplus}$ core can lose significant parts of its atmosphere in a timescale of tens of Myrs.

The process above paints a picture of a Neptune-like planet, which is migrating inwards due to tidal interactions, while simultaneously losing parts of its atmosphere on the way. It is currently unknown to what extent the loss of significant parts of an atmosphere will affect a planet’s tidal Love number. This would in turn affect the speed of tidal migration, and therefore this interplay could potentially have a large effect on both the orbital evolution, as well as the final size of a planet. It is therefore important to know the effect of a planet’s atmosphere on its tidal Love number, which is what will be explored in this study.

In order to do this, we adopted a simple model of a planet with a homogeneous rocky core with a homogeneous gaseous atmosphere. The ALMA3 code by Melini et al. (2022) is used to calculate the Love number for different sizes of the atmosphere, to see the effect of a planet losing its atmosphere on its Love number and the tidal migration it will undergo. In Sec. 2 we start by more thoroughly introducing (tidal) Love numbers and the ALMA3 code used for this study. In Sec. 3 we briefly discuss how the tidal Love number influences the migration rate. Sec. 4 will define the fiducial model adopted in this study. In Sec. 5 we discuss an analytical solution for the Love number of a fluid atmosphere which can be used to compare our results. We will then move on to the results and discuss them in Sec. 6. Finally we will present our conclusions in Sec. 7.

2 Tides & Love numbers

2.1 Tidal interactions

When taking into account the fact that, seen from the forcing body, the back side of the responding body is further away than the front side, and since gravity drops with the distance squared, different parts of the responding body experience different gravitational attraction from the forcing body. Due to these

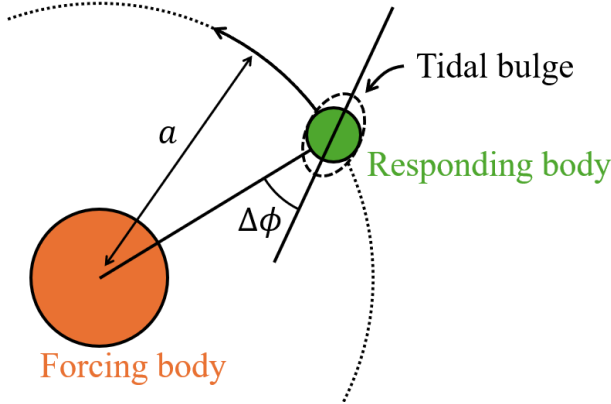


Figure 1: A responding body (green) on an orbit with semi-major axis a is perturbed by a forcing body (orange). The resulting tidal bulge is pointing away from the forcing body with a lag angle $\Delta\phi$.

differential forces, the responding body will be deformed until the self-gravity of the deformed body restores hydrodynamic equilibrium after some time. Depending on the internal structure of both bodies, the deformation can assume complex shapes. But to first order approximation, the deformation consists of two tidal bulges at which the responding body is stretched out. As the forcing body orbits the responding body, or the other way around, the tidal bulges rotate around the surface of the responding body. Due to the responding body resisting the deformation because of its internal rigidity, the tidal bulges cannot directly follow the forcing body's motion, but are instead delayed, and lag behind by an angle $\Delta\phi$, called the phase offset or lag angle (see Fig. 1). This phase offset will cause dissipation of energy to occur inside of the planet, heating up its interior. This energy is extracted from the rotation rates of both bodies and from the orbital energy, and can cause tidal migration.

2.2 Love number definition

The tidal Love number k relates the perturbing potential to the perturbed body's response. When a non-rigid body is perturbed by some external poten-

tial U , it will deform due to this. This deformation in turn causes the potential from the body's self-gravity Φ to change, so $\Phi \rightarrow \Phi + \delta\Phi$. The tidal Love number k is defined as the ratio of the change in self-gravity to the perturbing potential at the body's surface ($r = R$) (Storch and Lai, 2015), so

$$k \equiv \frac{\delta\Phi(R)}{U(R)}. \quad (1)$$

The body will deform due to the external perturbing potential, and eventually settle into an equilibrium again, in which case the perturbed potential must satisfy Poisson's equation (Storch and Lai, 2015)

$$\nabla^2\Phi = -\frac{Gm(r)}{r} \quad (2)$$

with G the gravitational constant and $m(r)$ the mass within radius r . Depending on the precise shape and density profiles of both bodies, the resulting potential response $\Delta\Phi$ can assume complex dependence on the radial position on the planet. This complex behaviour can be decomposed into spherical harmonics, which provide a complete basis for functions on a sphere (Tremaine, 2023), such that we get

$$\Phi = \sum_{l,m} A_{lm} Y_l^m(\theta, \phi) \quad (3)$$

with Y_l^m the spherical harmonic of degree l and order m . In exoplanet research, as a first order approximation, only the lowest-order part is taken into account, which is the harmonic Y_2^0 with $l = 2$ and $m = 0$. We can neglect all higher order contributions and focus solely on the tidal Love number associated to this part of the response, denoted by k_2 (Storch and Lai, 2015).

2.3 The fluid Love number

For a constant, non-changing perturbation the time it takes for the body to adjust and deform due to the perturbation can be neglected and the deformation becomes static. More general, if the period of the perturbation is much longer than the time needed by the body to adjust, the deformation will directly follow the perturbation, as if it did not resist at all. This

is equivalent to having a body with zero rigidity, since those do not resist the deformation. In these cases, the fluid Love number k_2^f is applicable, which is given by the Darwin-Radau equation Correia and Rodríguez (2013)

$$\frac{I_{zz}}{mR^2} = \frac{2}{3} \left(1 - \frac{2}{5} \sqrt{\frac{4 - k_2^f}{1 + k_2^f}} \right), \quad (4)$$

where I_{zz} is the moment of inertia around the axis of rotation (here defined to be the z -axis). This way the Love number is a measure for the internal mass distribution and structure of a body. For a homogeneous sphere $I_{zz} = \frac{2}{5}mR^2$ (see Appendix D, and the fluid Love number reduces to

$$k_2^f = \frac{3}{2}, \quad (5)$$

which is the value of the tidal Love number for any homogeneous sphere for which self-gravity is more important than rigidity (Tremaine, 2023).

2.4 The frequency-dependent Love number

However, due to non-synchronous orbits, eccentricities or inclinations, the perturbation is not constant, and can even have a timescale shorter than the time the body needs to respond. By introducing a rheological model to describe the temporal response of a visco-elastic body to a perturbation, the Love number becomes dependent on the angular frequency of deformation ω . For this study, we adopt the Maxwell rheology, one of the simplest rheological models, which is described by a spring (for stiffness/rigidity) with a damper (for damping/viscosity) connected in series (Roylance, 2001). This model is often used to model solids. Gasses and fluids can be modeled using a Newton rheology, which is simply a damper, and therefore has no stiffness. For a Maxwell rheology, the frequency dependent Love number is given by Melini et al. (2022)

$$k_2(\omega) = k_2^f \left[1 - \frac{\gamma^2}{1 + \gamma^2} \exp\left(-\frac{1}{\omega\tau'}\right) \right] \quad (6)$$

where

$$\gamma^2 \equiv \frac{19\mu}{2g\rho R} \quad (7)$$

with μ the body's rigidity, g its surface gravity, ρ its density, R its radius and

$$\tau' = (1 + \gamma^2)\tau \quad (8)$$

with

$$\tau \equiv \frac{\eta}{\mu} \quad (9)$$

the Maxwell relaxation time, which is a measure for the time it takes a solid with a so-called Maxwell rheology to adjust to an introduced perturbation, depending on its viscosity η and rigidity μ .

2.5 The elastic Love number

For the rocky layers of the Earth, $\mu \approx 10^{11}$ Pa, and $\eta \approx 10^{21}$ Pa s, therefore according to Eq. 9 the Maxwell relaxation time is $\tau \approx 10^{10}$ s ≈ 300 yrs. Since we are interested in either moons or planets in close orbits, the perturbation timescale is much smaller than this. In this case $\omega \gg 2\pi/\tau$, such that we can use $\exp -1/(\omega\tau') \approx 1$ reducing Eq. 6 to

$$\lim_{\omega \gg 1/\tau} k(\omega) = \frac{k_2^f}{1 + \gamma^2}. \quad (10)$$

This limit is called the elastic limit, and using Eq. 7 the elastic Love number can be written as

$$k_2^e = \frac{k_2^f}{1 + \frac{19\mu}{2g\rho R}}. \quad (11)$$

In the case of a homogeneous sphere $g = \frac{4\pi}{3}G\rho R$ and $k_2^f = 3/2$ so the elastic Love number is

$$k_2^e = \frac{3}{2 \left[1 + \frac{47\mu}{8G\rho^2 R^2} \right]}. \quad (12)$$

This model of a homogeneous incompressible sphere is also called a Kelvin model or Kelvin sphere Spada and Boschi (2006). These limiting equations provide some insight in how the Love number changes with varying parameters:

- Increasing the radius R or density ρ of the planet will increase the elastic Love number k_2^e , without affecting the fluid Love number.

- Increasing the rigidity μ will decrease the elastic Love number k_2^e as well as decreasing the Maxwell relaxation time τ and with that the cross-over between elastic and fluid Love numbers.
- Increasing the viscosity η will not change the fluid or elastic Love numbers, but only increase the Maxwell relaxation time τ .

These equations will be used in section 6.1 to benchmark the first ALMA3 results.

2.6 Quality factor

Another quantity often used in the study of tidal interactions is the quality factor Q . As a body is deformed by a forcing body, the gravitational force of the forcing body performs work on the interior of the perturbed body. The rate at which work is done varies over the perturbation period. Because of this periodicity, the amount of energy from the deformation that is stored inside of the perturbed body changes over time. The quality factor is proportional the ratio of the maximum amount of energy stored in the perturbed body over a single deformation period, denoted by W , to the amount of work done over a single period, denoted by ΔW (Tremaine, 2023). The expression for Q therefore reads

$$Q \equiv \frac{2\pi W}{\Delta W}. \quad (13)$$

The quality factor can also be expressed in terms of the phase offset of the perturbation, in which case it is given by

$$Q = \frac{1}{\sin(\Delta\phi)}, \quad (14)$$

with $\Delta\phi$ the lag angle (Tremaine, 2023). Therefore, the Love number k_2 and the quality factor Q are independent quantities used to quantify tidal interactions¹.

¹Sometimes complex valued Love numbers with an imaginary part are used. These also hold information about the lag angle. In that case the quality factor is embedded in the complex Love number (Melini et al., 2022).

2.7 ALMA3

ALMA3 (the plAnetary Love nuMbers cAlculator) is a code written by Melini et al. (2022) that calculates loading and tidal Love numbers for spherically symmetric planetary models consisting of multiple stacked spherical shells with given properties. It has been validated against semi-analytical solutions and benchmarked with other codes, and provides accurate results. For each shell the outer radius, density, rigidity and viscosity need to be defined, as well as a rheological model. Schematic representations of the available rheological models, including the Maxwell and Newton rheologies discussed above, are shown in Fig. 2. Furthermore, an inviscid fluid rheology can be used, and also an Andrade rheology (a more complicated combination of the rheologies in Fig. 2). For a review of different rheologies, we refer to Correia and Valente (2022) and Ragazzo and Ruiz (2016).

In order to run the ALMA3 code for multiple models efficiently, a Python wrapper was created for this project. Using this, ALMA3 can be passed arrays with parameter values for all layers of the planetary model, the model can be executed and the results loaded into Python for analysis. All code is available on GitHub².

3 Migration rate

Correia et al. (2011) gives the migration rate from tidal friction between two bodies:

$$\frac{\dot{a}_1}{a_1} = \sum_i \frac{2K_i}{\beta_1 a_1^2} \left[f_2(e_1) \cos \theta_i \frac{\omega_i}{n_1} - f_3(e_1) \right], \quad (15)$$

with a_1 the semi-major axis of the responding body, which is orbiting the forcing body, the dot indicates a time derivative. This is a sum of two contributions: first of all, the tidal dissipation in the central star ($i = 0$), as well as the dissipation in the body itself ($i = 1$). In the equation we have

$$K_i = \Delta t_i \frac{3k_{2i} G m_{1-i}^2 R_i^5}{a_1^6}. \quad (16)$$

²<https://github.com/slagnews/AtmosphereLoveNumbers>

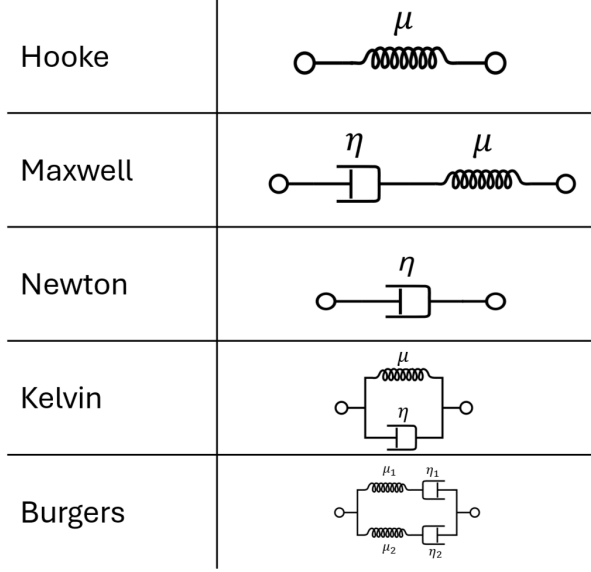


Figure 2: Different rheological models and their representation in terms of springs with a rigidity μ and dampers with viscosity η .

Furthermore $\beta_1 = (m_0 m_1)/(m_0 + m_1)$ is the reduced mass of the two-body system. ω_i is the rotation rate of the i th body around its own axis of rotation and n_i its mean-motion around the central body, given by $n_1 = \sqrt{\mu_1/a_1^3}$ with $\mu_1 \equiv G(m_0 + m_1)$. θ_i is the alignment angle between the axis of rotation of a body and the axis along which its tidal love number is defined. Lastly, Δt_i is the time lag between the tidal perturbation and the resulting distortion, which is given in terms of the lag angle (see Fig. 1) and the perturbing frequency ω by Remus et al. (2012) as

$$\Delta t = \omega \Delta \phi. \quad (17)$$

To get a rough estimate for the migration rate, we can make some simplifying assumptions. First of all, we can assume circular orbits, setting $e_1 = 0$ which yields $f_2(e_1) = f_3(e_1) = 1$. Furthermore we can assume that the axes of rotation and Love number are aligned, so we get $\cos \theta_i = 1$. This results

in the migration rate

$$\frac{\dot{a}_1}{a_1} = \sum_i \frac{2K_i}{\beta_1 a_1^2} \left[\frac{\omega_i}{n_1} - 1 \right]. \quad (18)$$

The direction of migration depends on the rotation rates ω_i of both bodies. When looking at the Earth-Sun system, the Earth rotates much faster around its axis than around the Sun, so $\omega_1/n_1 \approx 365$, causing outward migration. However the Sun rotates around its own axis in about a month, so $\omega_0/n_1 \approx 1/12$, causing inward migration. Looking at the Earth-Moon system, we see that the Moon is always facing the Earth with the same side, so $n_1 = \omega_1$, so there is no tidal dissipation in the Moon anymore, and all migration is caused by dissipation in the Earth. Since for the earth $\omega_0/n_1 \approx 30$, it causes the Moon to migrate outwards.

For this report, we consider planets that have low rotation rates compared to their orbit. Also, we are only interested in the tidal deformation of the planet itself, and will disregard the deformation and dissipation in the star. In this case, tidal dissipation in the planet will cause it to migrate inwards. In the limit $\omega_1 \ll n_1$ we get for the migration rate

$$\frac{\dot{a}}{a} \approx \frac{2K}{\beta a^2}, \quad (19)$$

where we removed the subscript for planet properties for brevity. Using the equations above, and $M_\star \gg M_p$, this evaluates to

$$\frac{\dot{a}}{a} \approx 6\Delta t k_2 R^5 \frac{GM_\star}{a^8}. \quad (20)$$

The tidal migration rate $\dot{a}/a$ depends on the planet's properties through the factor $\Delta t k_2 R^5$. For this study we will focus on the changes in the Love number, leaving the changes of the time lag Δt for further research.

4 Fiducial model

In order to study the effect of the atmosphere on the Love number, a two-layer fiducial model was adopted,

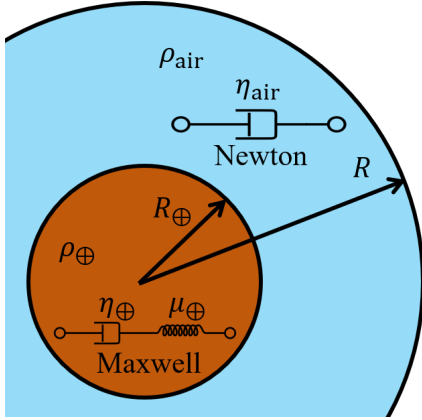


Figure 3: A schematic overview of the fiducial model adopted for this study. Both layers are homogeneous. The first layer represents a homogeneous rocky core with Maxwell-rheology with Earth-like density, viscosity and rigidity (see Tab. 2), extending up to 1 Earth radius. The second layer represents an atmosphere with Newton-rheology with viscosity and density equal to those of air at sea level, extending up to a variable outer radius R .

which can be seen in Fig. 3. The first layer is a homogeneous Earth-like core, using the average density of the Earth. A Maxwell rheology is used to model the rocky properties, with the viscosity and rigidity equal to those of the rocky layers in the Earth, adapted from the values used by Melini et al. (2022). The second layer represents the atmosphere, and has a Newton-rheology with density and viscosity equal to those of air at sea level (see Tab. 2 for all values and quantities used in this report). Since ALMA requires all parameters to be defined for each layer, the rigidity of the atmosphere is set to a small value of 1×10^{-7} Pa (negligible compared to the rigidity of the rocky layer, which is 1×10^{11} Pa), as suggested in a conversation with the authors of ALMA3, to avoid numerical instabilities but still model a non-rigid layer. The atmospheric layer extends up to a radius R , which will be varied from $R_{\oplus}$ (corresponding to a bare core) up to three times the radius of Jupiter, to see the effect of a changing atmosphere on the Love number.

5 Analytical derivation for polytropic atmosphere

Storch and Lai (2015) analytically calculated the Love number for a planet with a rocky core and a fluid envelope. Even though this study was specifically on gaseous atmospheres, the results provide an interesting comparison. They assumed a polytropic $n = 1$ equation of state $P = K\rho^2$ with K an arbitrary constant which, in hydrostatic equilibrium, results in a density profile given by

$$\rho(r) = \rho_0 \frac{\sin[q(1 - r/R)]}{qr/R} \quad (21)$$

with r the radial coordinate, R the planet's outer radius (including envelope), ρ_0 a reference density given by

$$\rho_0 \equiv \frac{q^2 M_p}{4\pi R^3} \quad (22)$$

with M_p the mass of the planet and

$$q^2 \equiv \frac{2\pi G R^2}{K}. \quad (23)$$

The density jump at the core surface from ρ_{core} inside the core to $\rho_{\text{max}} = \rho(R_{\text{core}})$ just outside the core, is constrained by

$$\frac{\rho_{\text{core}}}{\rho_{\text{max}}} = \frac{3R^2}{q^2 R_{\text{core}}^2} \left[\frac{q R_{\text{core}}}{R} \cot[q(1 - R_{\text{core}}/R)] + 1 \right]. \quad (24)$$

To arrive at the chosen R , R_{core} and M_p these equations are solved for the right value of K , which is a free parameter. In the original study, this model was used to calculate the Love number of a planet with a fixed outer radius and a changing core radius, with the goal of inferring the internal structure of an exoplanet based on an observed Love number. For our purpose, we shall use the same result to study how the Love number changes for a known core density with a varying atmosphere radius in Sec. 6.5.

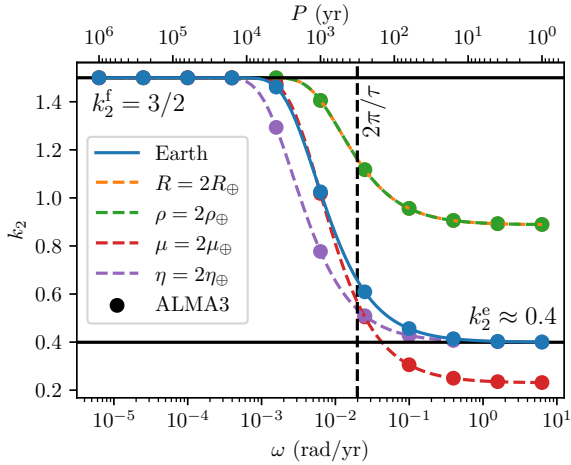


Figure 4: The Love number k_2 as a function of the deformation frequency ω (radians per year) and period P (years). The blue line indicates the result for a homogeneous Earth. The dashed lines correspond to models where one of the parameters was changed. The horizontal lines indicate the fluid and elastic limit of the Earth model. The vertical dashed line indicates the Maxwell period. The circles indicate results from ALMA3.

6 Results & Discussion

6.1 Frequency dependent Love number

As a first test of the ALMA3 code, we ran simulations on the fiducial model (blue) described in Sec. 4 to retrieve the time-dependent Love number, as can be seen in Fig. 4. We can see that the analytical solution from Eq. 6 (blue) lines up well with the retrieved Love numbers from ALMA3 (orange). For large ω we can see the Love number converging on the elastic Love number, and for small ω on the fluid Love number as expected. The change happens around the Maxwell relaxation time τ . We can also see what happens when we change the different parameters of the model. In Fig. 4 we compare the analytical results with the results from ALMA3 again, and see that the parameters change the Love number as expected

from Eq. 6: a larger radius R and higher density ρ both increase the elastic Love number (high ω , on the right) without changing the Fluid love number or the cross-over time. An increased rigidity μ decreases the elastic Love number as expected and, although not clearly visible because the elastic limit is lower, also shifts the cross-over frequency to the right. An increased viscosity η only changes the cross-over frequency, without changing the Love number in any of the two limits. All ALMA3 results match the analytical expectations precisely.

6.2 Varying atmosphere size

To quantify how atmospheric loss affects a planet's tidal Love number, ALMA3 was used to model planets with varying atmosphere sizes. Starting from a bare rocky core, the height of the atmosphere was gradually increased up to three Jupiter radii. This was done for various core radii. For each model, we computed the Love number for a deformation frequency of one per year, deep in the high-frequency elastic regime—typical for close-in exoplanets. The results, shown in Fig. 5, reveal three distinct regimes of tidal response depending on the atmospheric size relative to the core.

To understand these trends, we consider Eq. 20, which shows that the tidal migration rate scales with $k_2 R^5$ —a quantity we refer to as the *migration rate factor*. This factor captures the strength of the tidal response relevant for orbital evolution, and is shown in Fig. 6.

In the core-dominated regime, the atmosphere is small and contributes negligibly to the tidal dissipation. Therefore the migration rate factor will remain fairly constant as the atmosphere shrinks, such that $k_2 R^5 \approx k_2^{\text{core}} R_{\text{core}}^5$, where k_2^{core} is the tidal Love number of the homogeneous rocky core, which is given by Eq. 12. Therefore, the Love number increases as the atmosphere shrinks, but this is compensated for by the planet's decreasing outer radius, keeping the migration rate factor largely constant. This regime is valid for $R \lesssim 3R_{\text{core}}$.

In the atmosphere-dominated regime, the tidal dissipation is dominated by the atmosphere, and the whole planet acts as a homogeneous fluid sphere, for

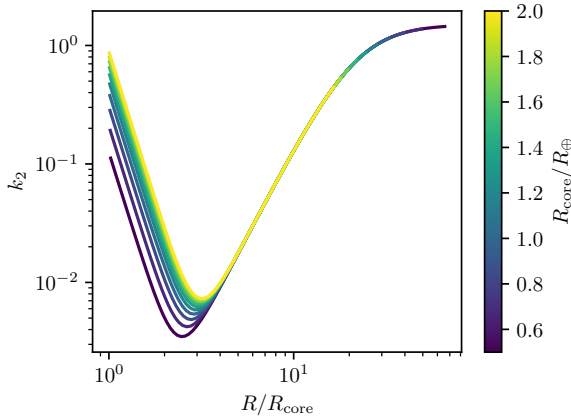


Figure 5: The Love number k_2 is plotted against the outer radius R of the planet relative to the planet’s core radius R_{core} for different core radii. All the way on the left $R = R_{\text{core}}$ and the planet has no atmosphere at all. The size of the atmosphere increases to the right. The Love number decreases at first, before increasing again and settling at a constant value of $3/2$.

which $k_2 = k_2^f = 3/2$ independent of size. Therefore, the Love number itself remains largely constant as the atmosphere shrinks. However, since the planet’s outer radius decreases, the migration rate factor does significantly decrease when a significant part of the atmosphere is lost. This regime is valid for $R \gtrsim 20R_{\text{core}}$.

In between these two regimes, when both core and atmosphere play a significant role in the planet’s tidal response, as the atmosphere radius increases the Love number decreases at first, before increasing and settling on a constant value.

6.3 Varying parameters

Above we showed how the Love number and migration rate factor change as the atmosphere changes in size for the fiducial model as well as for similar models with different core radii. Apart from the core radius R_{core} , the relevant parameters are the core density ρ_{core} , the atmosphere density ρ_{atm} and the core

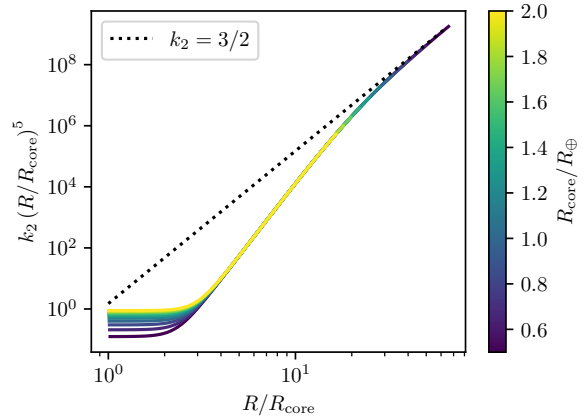


Figure 6: The migration rate factor $k_2 R^5$ is plotted against the outer radius R of the planet, relative to its core radius R_{core} , for various core radii. On the left $R = R_{\text{core}}$ and the planet has no atmosphere at all, the size of the atmosphere increases to the right. The migration rate factor is constant for small atmospheres, and settles on a power-law corresponding to $3/2 R^5$ for large atmospheres on the right.

rigidity μ_{core} . As shown above, the rigidity of the atmosphere is zero, and the viscosity of the core as well as of the atmosphere only change the cross-over point between the different regimes, without influencing the Love number in the elastic regime, which we are looking at here. It is therefore interesting how the behaviour of the Love number changes as these parameters change as well.

Fig. 7 shows the same results as Fig. 6, but now the density of the atmosphere is being varied. In our fiducial model, we assume a density equal to the density of the Earth’s atmosphere at sea level, here we can also see what happens when the atmosphere is up to 100 times as dense (corresponding to a density of 129 kg m^{-3}). The overall behaviour of the migration rate factor remains the same, but the point at which the atmosphere starts to become important, and where the migration rate factor starts to increase shifts to the left for a denser atmosphere. This makes intuitive sense, as a denser atmosphere already has a larger mass, and therefore a greater influence on the

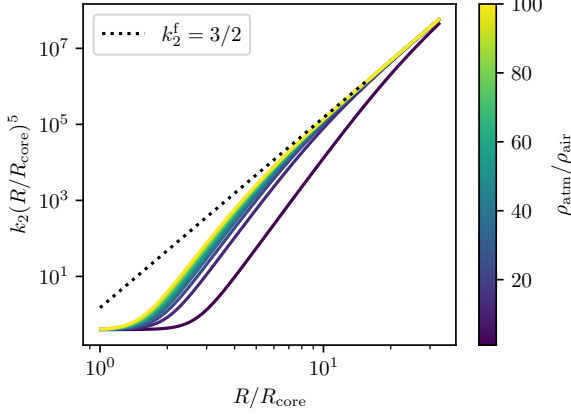


Figure 7: The migration rate factor $k_2 R^5$ is plotted against the outer radius R of the planet, relative to its core radius R_{core} , for various atmosphere densities ρ_{atm} . As the density of the atmosphere increases, the point at which the atmosphere becomes important and eventually dominant shifts to lower radii R . For any density the migration rate factor settles on a power-law corresponding to $3/2 R^5$ (dotted black line) for large atmospheres on the right.

deformation of the planet, at a smaller radius R . Fig. 7 again shows the same results as Fig. 6, but now the density of the core is being varied. An increased core density causes the point where the atmosphere becomes important to shift to higher values of R , as the relative mass and therefore influence of the atmosphere decreases when the core is denser. It also causes the Love number and migration rate factor to increase as the density increases for the $R = R_{\text{core}}$ limit, since for a bare core the Love number is given by Eq. 12, which increases with density. Finally, in Fig. 9 we show the effect of a varying core rigidity μ_{core} . We can see that an increased core rigidity decreases the Love number and migration rate factor in the $R = R_{\text{core}}$ limit where the planet has no atmosphere. This is in accordance with Eq. 12. The rigidity of the core does not seem to affect the size of the atmosphere needed for it to become important and eventually dominant for the Love number, as all models converge on the $\frac{3}{2} R^5$ asymptote at the same

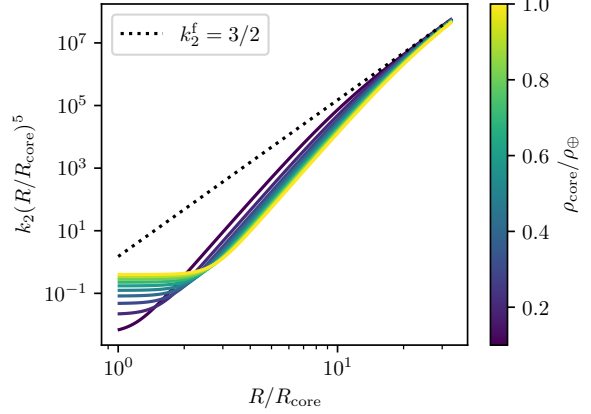


Figure 8: The migration rate factor $k_2 R^5$ is plotted against the outer radius R of the planet, relative to its core radius R_{core} , for various core densities ρ_{atm} . As the density of the core increases, the point at which the atmosphere becomes important and eventually dominant shifts to higher radii R . It also causes the value of the Love number and migration rate factor to increase in the limit where $R = R_{\text{core}}$, as this is where the planet has no atmosphere and the Love number is given by Eq. 12. For any density the migration rate factor settles on a power-law corresponding to $3/2 R^5$ (dotted black line) on the right.

time.

6.4 Analytical expression for k_2

The results presented above provide clear insight into the evolution of the Love number and migration rate during atmospheric loss. For practical use in planetary migration simulations (such as REBOUNDx and TidyMess), where k_2 is often treated as an input parameter, it is desirable to replace costly ALMA3 runs with a reliable analytical approximation Boekholt and Correia (2023); Tamayo et al. (2019).

Finding a suitable function to fit the results with proved to be difficult. Even though the limits of very small and very large atmospheres converge on constant values or simple power-laws, the behaviour in-between is less predictable. Various ideas were tried

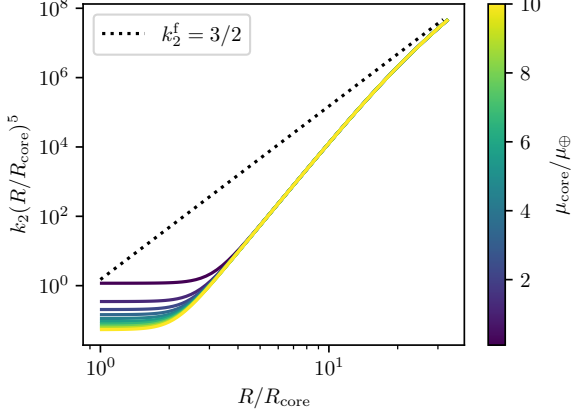


Figure 9: The migration rate factor $k_2 R^5$ is plotted against the outer radius R of the planet, relative to its core radius R_{core} , for various core rigidities μ_{core} . An increased core density causes the Love number and migration rate factor to decrease, in accordance with Eq. 12. It does not appear to change the point at which the atmosphere becomes important/dominant, as all models settle on the $\frac{3}{2}R^5$ asymptote (dotted black line) at the same point.

to find a good analytical relation, all described in Appendix B. In the end a relation derived from the Darwin-Radau equation (Eq. 4) proved to work well. The equation used is

$$k_2(R) = \frac{k_2^{\text{core}} + k_2^f \frac{2}{5} \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left[\left(\frac{R}{R_{\text{core}}} \right)^8 - 1 \right]}{1 + \frac{2}{5} \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left[\left(\frac{R}{R_{\text{core}}} \right)^3 - 1 \right]} \left(\frac{R}{R_{\text{core}}} \right)^{-5}. \quad (25)$$

This appears complicated at first, but it can be broken down for a deeper understanding. The overall structure of the equation can be seen as the ratio of the fifth moment of mass (a higher-order version of the moment of inertia, which is also known as the second moment of mass) to the mass itself (which can be seen as the zero'th moment of mass), divided over R^5 . In the limit where $R/R_{\text{core}} \approx 1$, so where the planet has no atmosphere, the equation simplifies to

a power-law, namely

$$\lim_{R/R_{\text{core}} \leftarrow 1} k_2(R) = k_2^{\text{core}} \left(\frac{R}{R_{\text{core}}} \right)^{-5}. \quad (26)$$

This is in correspondence with what we see in Fig. 5, and with the constant value of the migration rate factor on the left side in Fig. 6. In the opposite limit where $R/R_{\text{core}} \gg 1$, all powers of R/R_{core} cancel out, and it settles at a constant value

$$\lim_{R/R_{\text{core}} \rightarrow \infty} k_2(R) = k_2^f, \quad (27)$$

which is also in accordance with the horizontal asymptote on the right side of Fig. 5, and the power-law asymptote that the migration rate factor settles on in Fig. 6.

From analytical inspection, the function was thought to be a good approximation to the Love number. In Fig. 10 the result from ALMA3 for the fiducial model is compared to this analytical function, using $k_2^f = 3/2$ and Eq. 12 for k_2^{core} .

As the analytical expression fits well for the fiducial model, and since this model used arbitrarily chosen parameter values, it is to be expected that this expression will also generalize to other parameter values. Each of the relevant parameters was changed to another value, while keeping all other parameters at their fiducial values, and the results from ALMA3 compared to analytical result in Fig. 11. As all errors are well below 1%, the analytical relation generalizes well to other parameter values.

6.5 Comparison with fluid atmosphere model

As described in Sec. 5, Storch and Lai (2015) created an analytical model for the tidal Love number of a planet with a fluid atmosphere in hydrostatic equilibrium. This provides an interesting comparison with the results obtained using ALMA3. Therefore the found analytical solution was implemented and root-finding was used to find the value of K to satisfy the constraints on the core radius, core density and atmosphere density. To validate the implementation, Fig. 1 from Storch and Lai (2015) was recreated, and

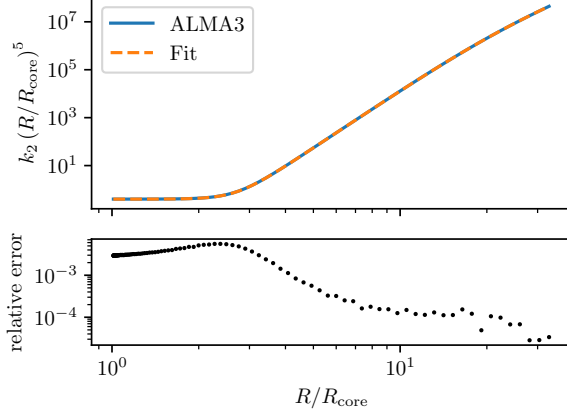


Figure 10: The migration-rate factor $k_2 (R/R_{\text{core}})^5$ against the size of the planet relative to its core R/R_{core} for an Earth-like planet with an air-like atmosphere. The results from ALMA3 (blue) are compared to the analytical expression in Eq. 25 with the function described in Eq. 25 (orange). The relative error between the two is shown below.

can be seen in Fig. 12, showing how k_2 changes as the core radius is changed, keeping the outer radius of the planet constant. Since the resulting figure is exactly the same as the original from their paper, the implementation must be correct, and can be used to study how the Love number changes as the atmosphere radius changes.

By keeping the core radius constant and varying the atmosphere radius in this analytical model, the resulting values for the Love number k_2 and migration rate factor $k_2 R^5$ can be compared. The result can be seen in Fig. 13.

The results shown here are not for the fiducial model used before, but for a planet with a homogeneous rocky with the same radius and mass as the entire planet Neptune, and an atmosphere with the density of air at sea level on Earth. Since the solution to the equations from Storch and Lai (2015) described in Sec. 5 behave poorly for extremely small atmospheres (at $R \approx R_{\text{core}}$ the atmosphere is very small, and the solution for the equation of state parameter K becomes unstable), large ratios between the core

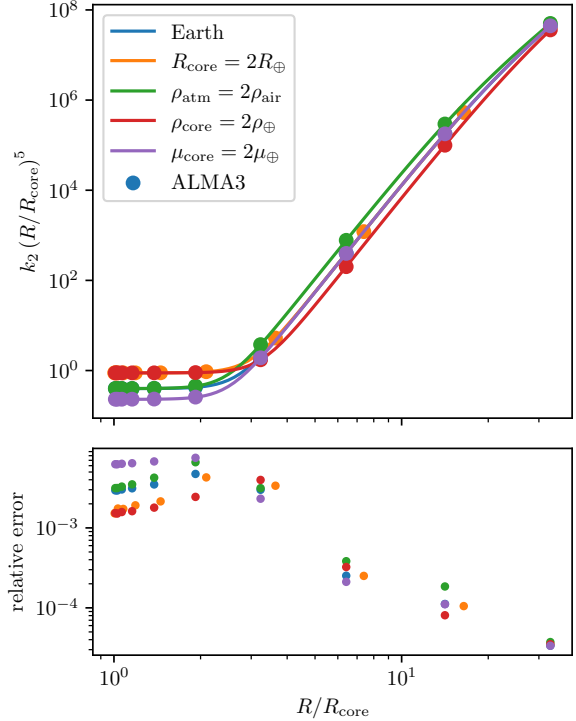


Figure 11: The migration-rate factor $k_2 (R/R_{\text{core}})^5$ against the size of the planet relative to its core R/R_{core} for the fiducial ("Earth"-like) model and models where a single indicated parameter was changed. The results from ALMA3 (dots) are compared to the analytical expression in Eq. 25 the function described in Eq. 25 (lines). The relative error between the two is shown below.

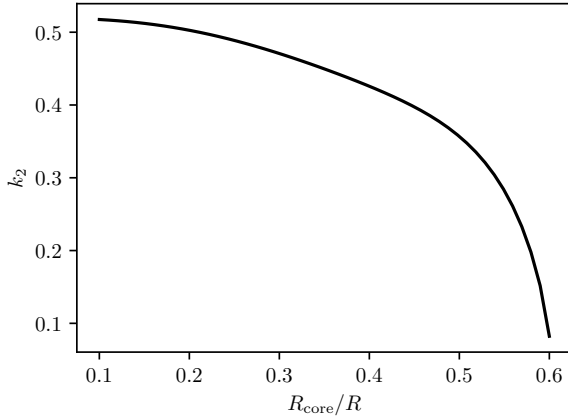


Figure 12: The tidal Love number k_2 as a function of the core radius R_{core} relative to the planet's outer radius $R = R_{\text{Jup}}$. The planet's mass is fixed at $1 M_{\text{Jup}}$, it has a rocky core and a fluid atmosphere. The result matches the result in Fig. 1 in Storch and Lai (2015).

density and atmosphere density were not feasible.

For the ALMA3 model of this fluid atmosphere, the amount of shells that can be used to model the density profile of the atmosphere is limited by computational time and stability. To keep the simulation of more than 3 layers stable, a higher simulation precision (Gaver sequence order and the number of digits) is needed. This however significantly slows down the simulation by a factor of 10 or more. Therefore 10 layers were used, which provided a stable results at acceptable simulation speeds. The results appear to be in close agreement, particularly for large atmosphere radii. Both approach the $k_2 = 3/2$ asymptote, even though the result from the analytical model does not quite reach it, but remains lower. In this limit, the atmosphere is much larger than the core, and the planet acts as a completely fluid sphere. The discrepancy can be explained using the Darwin-Radau equation (Eq. 4), which states that the fluid Love number increases as the moment of inertia of a planet increases. Since the analytical model by Storch and Lai (2015) has a density distribution described by Eq. 21, it has a higher concentration of mass in the

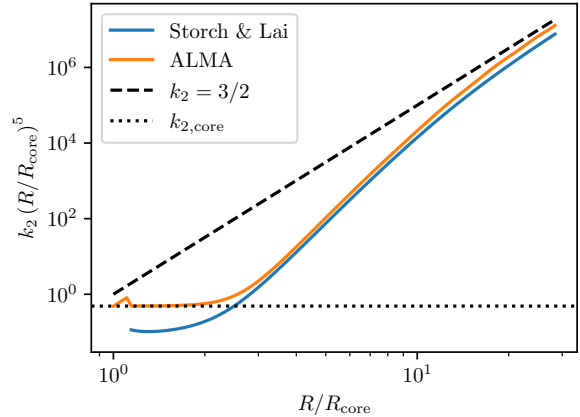


Figure 13: The migration rate factor $k_2 R^5$ as a function of the planet's outer radius R relative to its core radius R_{core} . The model used is a planet with a rocky core and a fluid envelope in hydrostatic equilibrium. The results from ALMA3 simulations (blue) are compared to the results from the analytical derivation by Storch and Lai (2015) (orange).

center, resulting in a lower moment of inertia, and therefore a lower fluid Love number.

The values from the two methods in Fig. 13 do not agree in the limit of very small atmospheres, even though this was to be expected, since both should converge on the same Love number corresponding to a bare core as both models use the same core size. The discrepancy for small atmospheres is thought to be due to the fact that the analytical model by Storch and Lai (2015) assumes there to be some atmosphere, and some parameters, such as the constant K in the equation of state, are set such that certain boundary conditions are met. As $R \rightarrow R_{\text{core}}$, the atmosphere becomes extremely small, which might break some of these boundary conditions and blow up values such as K , leading to large errors.

7 Conclusions

This study explored how the loss of a planetary atmosphere impacts the tidal Love number and the re-

sulting tidal migration rate. The main conclusions are summarized below:

- The impact of atmospheric escape on the Love number and tidal migration rate was shown to depend strongly on the nature of the planet:
 - **Atmosphere-dominated planets** behave almost like homogeneous fluid spheres, for which $k_2 = 3/2$, independent of density and radius. Loss of the atmosphere will therefore leave the Love number mostly unchanged, but the decrease in planetary radius causes a strong decrease in tidal migration rate. Potentially altering the final position and size of such planets.
 - **Planets with thin atmospheres** respond to tidal perturbations as a rigid, core-dominated body, loss of parts of the thin atmosphere cause the tidal Love number k_2 to decrease sharply. The decreasing planetary radius cancels this effect, leaving the tidal migration rate largely unchanged, as most of the tidal dissipation already happened in the core, which remains unchanged.
- **Insensitivity to viscosity:** The viscosities of the core and atmosphere influence the perturbation frequency at which the Love number transitions from fluid to elastic response but do not affect the Love number in the high-frequency (elastic) regime, which is important for close-in planets.
- **Analytical fit:** The behaviour of the Love number across varying atmosphere sizes is accurately captured by the analytical expression derived in the study (Eq. 25), enabling fast integration into tidal evolution models without requiring full numerical simulations in future studies.
- **Phase lag angle:** The effect of atmospheric escape on the phase lag angle $\Delta\phi$, which indicates how far the deformation lags behind the perturbation, was not investigated in this study. Since the tidal migration rate is directly proportional to this quantity, the effect of atmospheric escape

on the phase lag angle should be investigated in future studies.

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A Spherical Coordinates

For this report, we adapt the convention for spherical coordinates that has

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} r \sin \theta \cos \phi \\ r \sin \theta \sin \phi \\ r \cos \theta \end{pmatrix}. \quad (28)$$

In this case, a volume-element dV is given by

$$dV = r^2 \sin \theta dr d\theta d\phi. \quad (29)$$

Furthermore, the squared distance to the z -axis is given by

$$x^2 + y^2 = r^2 \sin^2 \theta. \quad (30)$$

B The quest for $k_2 = f(R)$

The main goal of this research was to characterize how the Love number k_2 changes with a changing atmosphere size R . By fitting the simulation results with an analytical function, this fit can be used as input for orbital migration simulations, which would remove the need to run ALMA3 simulations to update k_2 during these migration simulations. In order to get a good fit, a suitable analytical function $k_2(R)$ needs to be found. In order to do this, we switch to dimensionless parameters, where we adopt

$$x \equiv \frac{R}{R_{\text{core}}} \quad (31)$$

$$\rho \equiv \frac{\rho_{\text{atm}}}{\rho_{\text{core}}}. \quad (32)$$

Since we are varying the size of the atmosphere, and the atmosphere is put on top of the core, $x > 1$. Furthermore, $\rho \ll 1$ since the atmosphere is much less dense than the core. Below we discuss how several functions have been found that can be used to fit the behaviour of the Love number.

B.1 Derivation from Darwin-Radau

The Darwin-Radau equation gives an expression for the fluid Love number. Since we are adding a non-rigid atmosphere to our rigid core, the first-order

approximation to the limit with a large atmosphere should be the fluid Love number of a homogeneous sphere, which is $k_2^f = 3/2$. A second-order approximation would be to still use the Darwin-Radau equation, but take into account that the planet is made up of two layers with differing density, changing the moment of inertia I_{zz} . This moment of inertia is given by

$$I_{zz} = \iiint_V (x^2 + y^2) dm, \quad (33)$$

where $x^2 + y^2$ is the square of the distance to the z -axis. Using $dm = \rho dV$ and converting to spherical coordinates (see Appendix A for the used conventions, this reduces to

$$I_{zz} = \int_0^{2\pi} \int_0^\pi \int_0^R \rho(r, \theta, \phi) r^4 \sin^3 \theta dr d\theta d\phi. \quad (34)$$

For a spherically symmetric density, this reduces to

$$I_{zz} = \frac{8\pi}{3} \int_0^R \rho(r) r^4 dr. \quad (35)$$

In the case of our two-layer model, we get

$$I_{zz} = \frac{8\pi}{15} [\rho_{\text{core}} R_{\text{core}}^5 + \rho_{\text{atm}} (R^5 - R_{\text{core}}^5)], \quad (36)$$

which we can rewrite as

$$I_{zz} = \frac{8\pi}{15} \rho_{\text{core}} R_{\text{core}}^5 \left[1 + \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left(\left(\frac{R}{R_{\text{core}}} \right)^5 - 1 \right) \right] \quad (37)$$

$$= \frac{8\pi}{15} \rho_{\text{core}} R_{\text{core}}^5 [1 + \rho (x^5 - 1)]. \quad (38)$$

From the Darwin-Radau equation, we can see that the Love number depends on the unitless number I_{zz}/mR^2 , with m the total mass of the planet and R the outer radius. The total mass of the planet can be written in a similar way as I_{zz} , namely

$$m = \frac{4\pi}{3} \rho_{\text{core}} R_{\text{core}}^3 \left[1 + \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left(\left(\frac{R}{R_{\text{core}}} \right)^3 - 1 \right) \right] \quad (39)$$

$$= \frac{4\pi}{3} \rho_{\text{core}} R_{\text{core}}^3 [1 + \rho (x^3 - 1)]. \quad (40)$$

Using this, the unitless number evaluates to

$$\frac{I_{zz}}{mR^2} = \frac{2}{5}x^{-2} \frac{1 + \rho(x^5 - 1)}{1 + \rho(x^3 - 1)} \quad (41)$$

Removing the first factor, we get a function that changes with the radius

$$f_1(x) \equiv \frac{1 + \rho(x^5 - 1)}{1 + \rho(x^3 - 1)} \quad (42)$$

This function is plotted in Fig. 14. We can see that the behaviour of this function is very similar to how $k_2(R/R_{\text{core}})^5$ behaves as a function of R : it starts at a constant value at small x , then grows fast, after which the slope decreases a bit and it settles on a power-law asymptote. In this case the asymptote is $y = x^2$, whereas $k_2(R/R_{\text{core}})^5$ approaches $(R/R_{\text{core}})^5$ in the $R \gg R_{\text{core}}$ limit. This we can fix by changing the exponent of x in the numerator to 8, as $x \gg 1$, the function will approach $x^8/x^3 = x^5$. Furthermore, we want it to converge to $k_2^f(R/R_{\text{core}})^5$, which we can fix by adding the k_2^f as a factor as well. We now have the following function

$$f_2(x) = \frac{1 + k_2^f \rho(x^8 - 1)}{1 + \rho(x^3 - 1)}. \quad (43)$$

This behaves as we want in the $x \gg 1$ limit, but it does not converge on the correct value for small x . When x approaches zero ($R \rightarrow R_{\text{core}}$, i.e. a bare core), the Love number should converge on the value for a bare core k_2^{core} . This we can enforce by adding a factor to the constant value in the numerator, after which we get

$$f_3(x) = \frac{k_2^{\text{core}} + k_2^f \rho(x^8 - 1)}{1 + \rho(x^3 - 1)}. \quad (44)$$

After fitting with this function, it was found that it did not quite work yet. To see what should be changed, ρ , k_2^f and k_2^{core} were left as free parameters, and SciPy's curve fit was used to find the optimal parameters. As it turns out using $\frac{2}{5}\rho$ instead of ρ consistently resulted in almost perfect fits. Therefore the fitting function is defined as

$$f_3(x) = \frac{k_2^{\text{core}} + \frac{2}{5}k_2^f \rho(x^8 - 1)}{1 + \rho(x^3 - 1)}. \quad (45)$$

Now this function was used to fit $k_2(R/R_{\text{core}})^5$, so in order to fit k_2 itself, this factor has to be compensated for, and the final function to fit k_2 evaluates to

$$k_2(R) = \frac{k_2^{\text{core}} + k_2^f \frac{2}{5} \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left[\left(\frac{R}{R_{\text{core}}} \right)^8 - 1 \right]}{1 + \frac{2}{5} \frac{\rho_{\text{atm}}}{\rho_{\text{core}}} \left[\left(\frac{R}{R_{\text{core}}} \right)^3 - 1 \right]} \left(\frac{R}{R_{\text{core}}} \right)^{-5} \quad (46)$$

where we used Eqs. 31 and 32 to switch back from dimensionless to normal parameters.

It should be noted that even though this equation was inspired from the Darwin-Radau equation which is for *fluid* Love numbers, the fit is applied to the *elastic* Love number of a two-layer planet. This sounds contradictory at first, but there are two things at play here that might make it understandable: first of all, the equation was specifically modified to converge on the elastic Love number for a bare core in the regime for a small atmosphere. Secondly, since we are adding an atmosphere without rigidity, this behaves as a fluid, making it logical for the changes in the Love number to be reminiscent of how the fluid Love number behaves.

B.2 Bode plots

The relation between the planet's outer radius and $k_2 R^5$ is reminiscent of three consecutive power-laws with smooth cross-overs. These functions are also a common occurrence in Bode plots in signal analysis (Oppenheim et al., 1997). From this, a possible function to use is

$$y = \left(1 + \left(\frac{x}{x_1} \right) \right)^{a_1} \left(1 + \left(\frac{x}{x_2} \right) \right)^{a_2 - a_1}. \quad (47)$$

When $x_2 > x_1$, then this function assumes a constant value of 1 in the limit where $x \ll x_1$. In the right-hand limit, when $x \gg x_2$, the function becomes $y \propto x^{a_2}$. In between, so when $x_1 \ll x \ll x_2$, the function becomes $y \propto x^{a_1}$. Therefore, a_1 and a_2 are the slopes of the function, and x_1 and x_2 the cross-over points. However, the cross-over widths of this function are fixed, and not necessarily the same as in the results described above. This can be fixed by introducing

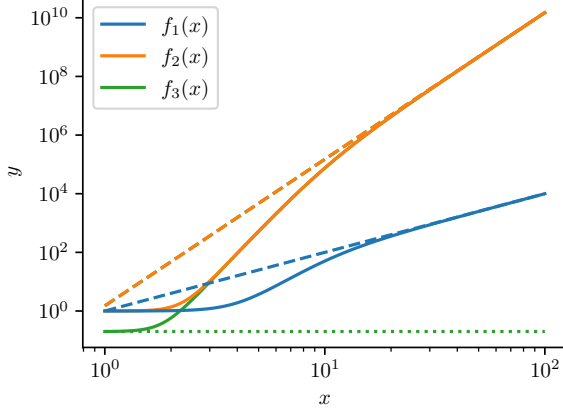


Figure 14: The functions $f_1(x)$ through $f_3(x)$ derived from the Darwin-Radau equation to fit the Love number. The behaviour is similar to the behaviour of $k_2(R/R_{\text{core}})^5$. From f_1 to f_2 the asymptote on the right was changed from x^2 to x^5 . From f_2 to f_3 the asymptote on the left was changed such that it can converge on the core Love number.

two extra parameters for the cross-over widths, w_1 and w_2 , and defining the function as

$$y(x) = \left(1 + \left(\frac{x}{x_1}\right)^{\sqrt{a_1 w_1}}\right)^{\sqrt{\frac{a_1}{w_1}}} \quad (48)$$

$$\left(1 + \left(\frac{x}{x_2}\right)^{\sqrt{(a_1 - a_2) w_2}}\right)^{-\sqrt{\frac{a_1 - a_2}{w_2}}} \quad (49)$$

As the widths w_1 and w_2 cancel themselves out when $x/x_1 \gg 1$ and $x/x_2 \gg 1$ respectively, the limiting behaviour of the function is conserved, but the cross-over widths can now be varied. From the results above, we want the function to converge on k_2^{core} on the left, and on $k_2^f(R/R_{\text{core}})^5$ on the right. Therefore we need to introduce some offsets, such that our final fit function becomes

$$k_2(x)x^5 = k_2^{\text{core}} + k_2^f[y(x) - 1], \quad (50)$$

with $x \equiv R/R_{\text{core}}$. Now in the case where $x \ll x_1$, $y(x) \approx 1$, so $k_2(x) = k_2^{\text{core}}$. In the case where $x \gg x_2$,

$y(x) \approx x^{a_2}/(x_1^{a_1} x_2^{-a_1+a_2})$. So if we require

$$x_1^{a_1} x_2^{-a_1+a_2} = 1 \quad (51)$$

or, equivalently,

$$a_1 = -a_2 \frac{\log x_2}{\log x_1/x_2}, \quad (52)$$

then $k_2(x) = k_2^f x^5$ for $x \gg x_2$. This means that the limit for a bare core, and the limit for an atmosphere-dominated planet are satisfied. The six parameters are the limiting Love numbers k_2^f and k_2^{core} , the cross-over points x_1 and x_2 the slopes a_1 and a_2 and finally the cross-over widths w_1 and w_2 . The limiting Love numbers are known, namely the core Love number given by Eq. 12 and $k_2^f = 3/2$ in the fluid limit. Furthermore the second slope $a_2 = 5$. The two cross-over points and the two cross-over widths are left as free parameters.

B.3 Effective rigidity

Another method that we tried to find a suitable function to fit the Love number was by starting from Eq. 12 for the elastic Love number of a Kelvin sphere. By using the average density

$$\bar{\rho} = \frac{3m}{4\pi R^3} \quad (53)$$

and the outer radius R we can define some "effective rigidity" $\bar{\mu}$ as

$$\bar{\mu} \equiv \frac{8G\rho^2 R^2}{47} \left(\frac{3}{2} \frac{1}{k_2} - 1\right). \quad (54)$$

By now using the results from the ALMA3 for k_2 simulations to reverse-engineer the effective rigidity. The result is visible in Fig. 15. As visible, this does not appear to be some easily fittable function as hoped for. Multiple fits were applied to this, among others an average of the rigidities of the core and atmosphere (which is non-rigid so effectively zero) weighted to size and mass. However, none of these fits proved succesful.

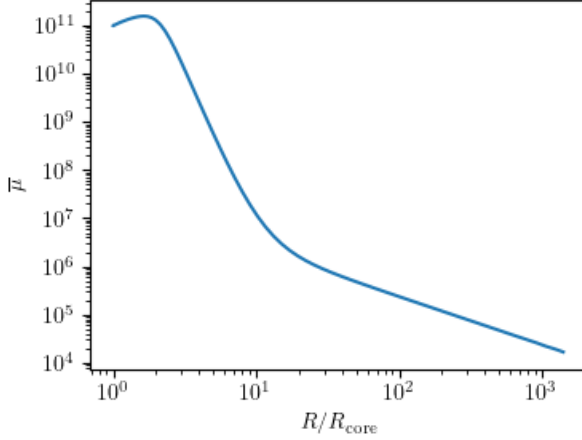


Figure 15: The effective rigidity $\bar{\mu}$ as defined in Eq. 54 as a function of the atmosphere radius R relative to the core radius R_{core} for the fiducial model. No suitable equation was found to properly fit this result.

Table 1: Used variables and their symbols.

Variable	Symbol	Unit
Love number	k_2	-
Fluid Love number	k_2^f	-
Elastic Love number	k_2^e	-
Core Love number	k_2^{core}	-
Outer planet radius	R	m
Core radius	R_{core}	m
Density	ρ	kg m ⁻³
Rigidity	μ	Pa
Dynamic viscosity	η	Pa s
Maxwell-relaxation time	τ	s
Deformation period	P	s
Deformation angular frequency	ω	rad yr ⁻¹
Gravitational acceleration	g	m s ⁻¹

Table 2: Used quantities, their symbols and values.

Quantity	Symbol	Value
Radius of the Earth	$R_{\oplus}$	6.371×10^6 m
Radius of Neptune	R_{Nep}	2.46×10^7 m
Radius of Jupiter	R_{Jup}	6.99×10^7 m
Mass of the Earth	$M_{\oplus}$	5.972×10^{24} kg
Mass of Neptune	M_{Nep}	1.024×10^{26} kg
Mass of Jupiter	M_{Jup}	1.90×10^{27} kg
Rigidity of the Earth	$\mu_{\oplus}$	10^{11} Pa
Viscosity of the Earth	$\eta_{\oplus}$	10^{21} Pa s
Gravitational constant	G	$6.67 \times 10^{-11} \frac{\text{m}^3}{\text{kg s}^2}$
Density of air	ρ_{air}	1.293 kg m^{-3}

C Used symbols & quantities

D Moment of inertia

The moment of inertia along the z axis of a body is given by

$$I_{zz} = \iiint (x^2 + y^2) dm \quad (55)$$

according to J. Peraire (2008). Using $dm = \rho r^2 \sin \theta dr d\theta d\phi$ in spherical coordinates, and $x^2 + y^2 = r^2 \sin^2 \theta$ we get

$$I_{zz} = \iiint \rho r^4 \sin^3 \theta d\theta d\phi dr. \quad (56)$$

For a radially symmetric sphere we have $\pi/2 < \theta < \pi/2$, $-\pi < \phi < \pi$ and $0 < r < R$ with R the outer radius, so the integral becomes

$$I_{zz} = \frac{8\pi}{3} \int_0^R \rho(r) r^4 dr \quad (57)$$

For a homogeneous sphere $\rho(r)$ is a constant and we get

$$I_{zz} = \frac{8\pi}{15} \rho R^5. \quad (58)$$

Using $M = \frac{4\pi}{3} \rho R^3$ this evaluates to

$$I_{zz} = \frac{2}{5} m R^2. \quad (59)$$

For a two-layered sphere, the integral can be split up and we get

$$I_{zz} = \frac{8\pi}{15} [\rho_1 R_1^5 + \rho_2 (R_2^5 - R_1^5)] . \quad (60)$$